

Course: Database Management System

Course Code: TBD

Credit:4

Course Description:

The course —*Database Management System* deals with a detailed study of principles of RDBMS and NoSQL databases along with the practical knowledge of database packages such as PostgreSQL and MongoDB.

Prerequisite(s): The student must know about a typical file system, data types like integer, float, and string, basic computer arithmetic, venn diagram representation of union, intersection, and complement of sets.

Note(s): Syllabus changes yearly and may be modified during the term itself, depending on the circumstances. However, students will be evaluated only on the basis of topics covered in the course.

Course Objectives:

- Introduce basic concepts of the database approach, the underlying models, and organizational issues.
- Learn about the relational database model as a logical view of data.
- Develop proficiency in data modeling.
- Explain the theoretical underpinnings of the relational database, including concepts like functional dependence, entity integrity, and relational integrity.
- Learn how a flawed data model can impact relational database implementation and manipulation.
- Learn about the relational database operators, the data dictionary, and the system catalog.
- Apply various relational algebra operations that provide the basis for relational database manipulation.
- Apply SQL operators to query and manipulate databases.
- Apply the query processing and optimization concepts to improve existing database design and query design.
- Learn about concurrency control and locking protocols.
- Learn concepts and usefulness of NoSQL databases.

Course Outcomes:

- Able to interpret the modeling symbols for the most popular ER modeling tools.
- Able to model the RDBMS schema with the help of ER models given a problem statement in English.
- Able to construct queries in SQL to manipulate live RDBMS.

- Able to analyze database requirements and determine the entities involved in the system and their relationship to one another in sophisticated database applications.
- Able to arrive at optimized queries and design database schemas for optimal performance.
- Choose appropriate concurrency control mechanism given the set of user requirements and constraints.
- Work with PostgreSQL DBMS in creating and manipulating data using simple queries and long transactions.
- Analyse the performance of queries.
- Choose between SQL and NoSQL database based on a given set of requirements.
- Use MongoDB for data storage, retrieval.

Approximate weightage of different components in evaluation:

Module Tests	20%
Lab Exam	20%
Midterm Exam	20%
Final Exam	40%

Course Policies:

General

1. Computing devices are not to be used during any exams unless instructed to do so.
2. Quizzes and exams are closed books and closed notes or as specifically announced by the instructor.
3. Quizzes are unannounced but they are frequently held after a topic has been covered.
4. Lab exam would be conducted without the availability of any network connectivity during the exam.
5. ***No makeup quizzes or exams will be given.***

Attendance and Absences

1. Attendance is expected and will be taken in each class. Students are not supposed to miss class without prior notice/permission. Any absences may result in point and/or grade deductions.
2. Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

Textbooks(s):

1. Fundamentals of Database Systems, 3rd ed., Addison-Wesley, 1998
R. A. Elmasri and S. B. Navathe
2. Database System Concepts, McGraw Hill, New Delhi, 1997
H. F. Korth and A. Silberschatz
3. Database Management Systems, 2nd ed., McGraw Hill, New York, 1999
R. Ramakrishnan and J Gehrke
4. An Introduction to Database Systems, Pearson Education, Eighth Edition, 2009
C. J. Date, A. Kannan and S. Swamynathan
5. Principles of Database Systems, Computer Science Press; 2nd edition (December 1982)
J D Ullman
6. PostgreSQL Documentation
URL: <https://www.postgresql.org/docs/>
7. MongoDB Documentation
URL: <https://www.mongodb.com/try/download/community>

Course Outline (tentative) and Syllabus:

The weekly coverage might change as it depends on the progress of the class. Each week assumes 4.5 hour lectures (inclusive of practicals as and when required). Quizzes will be unannounced, so students should maintain close to 100% attendance. There will be a quiz every week.

Week	Content
Week 1	• Database system concepts, three-schema architecture, data independence, database administrator, database user, client/server architecture • E-R diagram, mapping constraints, keys, generalization, Aggregation • Readings: Chapters 1,2 (EN)
Week 2	• Structure of relational database, relational data model and constraints, entity integrity, referential integrity, foreign Keys • Reducing E-R diagram to tables • Readings: Chapters 3,4,9 (EN)
Week 3	• Query languages, SQL programming, views • Readings: Chapters 6,7 (EN), 3,4,5 (SK), PostgreSQL doc
Week 4	• Relational algebra and relational calculus • Readings: Chapters 8 (EN)
Week 5	• Functional dependencies, normalization of relational databases - 1 NF, 2 NF, 3 NF, BCNF • Readings: Chapters 14 (EN)
Week 6	• Multivalued dependencies and 4 NF, Join dependencies and 5 NF, closure of a set of functional dependencies, canonical cover, closure of attribute sets, lossless decomposition, dependency preservation • Readings: Chapters 15 (EN)
Week 7	• File organization, sequential file, buffering, static hash Functions, dynamic hash functions, multilevel indexes, B+ tree index files, B tree index files • Readings: Chapters 16,17 (EN), 9 (RG)
Week 8	• Query processing • Readings: Chapters 18 (EN)
Week 9	• Query optimization • Readings: Chapters 19 (EN)
Week 10	• Transaction, properties of transaction, database recovery, shadow paging, recoverable schedule, serializable schedule • Readings: Chapters 20 (EN), 17(SK)
Week 11	• Lock-based protocol, timestamp-based protocol, multiple granularity, multiversion schemes, deadlock handling • Readings: Chapters 21 (EN), 18(SK)

Week 12	• Introduction to database security issues, discretionary access control based on granting and revoking privileges, mandatory access control and role-based access control for multilevel security, introduction to statistical database security, introduction to flow control • Readings: Chapters 30 (EN)
Week 13	• Introduction to MongoDB, operations • Data modeling, querying, indexing • Readings: MongoDB documentation